

Essential Insights: Week 2.1



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Predictive AI vs Generative AI

Predictive AI: Analyses historical data to estimate outcomes

- Risk scores, forecasts, likelihoods
- Supports human decision-making (e.g., insurance claims, dynamic pricing, predictive maintenance)

Generative AI: Creates new, usable content

- Documents, summaries, emails, code, memos
- Produces first drafts that professionals can review and refine

Key Example: Legal teams at Allen & Overy (2023) – 3,500 lawyers, 40,000 queries, widespread daily use for drafting client memos

What “Generation” Really Means

Generative AI produces artefacts, not just predictions - Creates structured business content ready for use:

- One-page regulatory summaries
- Variance explanations for finance
- Supplier follow-up emails
- First-draft legal advisory notes

Powered by **Large Language Models (LLMs)** based on Transformer architecture

Output is probabilistic → **Human review is essential** for accuracy and accountability

Autoregressive Generation – The Next Token Principle

How Generative AI Actually Builds Output

- Generates **one token at a time** based on everything produced so far
- **Token**: Small unit of text (word, sub-word, punctuation, number)
- Process = **Next-token prediction** (same principle as Gmail Smart Compose)
- No pre-planned answer – just sequential probability calculations
- Creates the illusion of reasoning through scale and context

Why Outputs Vary – Probabilistic Nature

Advantages:

- Rapid variation (multiple campaign messages, alternative descriptions)
- Creative exploration without starting from scratch

Risks:

- **Hallucinations** – plausible but incorrect statements
- Same prompt can produce different results
- Limited by **context window** (model “forgets” information outside the window)

Business takeaway: Treat outputs as strong first drafts, not final answers

Tokenisation – The Hidden Driver of Cost & Performance

Tokens are not the same as words

- 1 token $\approx$ $\frac{3}{4}$ of a word in English (varies with terminology)
- Complex business terms (e.g., “hyperautomation”, Indian names, acronyms) often split into multiple tokens

Directly impacts:

- Cost (input + output tokens)
- Speed/latency
- Context window size (model’s working memory)

Practical tip: Use clear, structured prompts and summarise inputs to control token usage

From Tokens to Meaning – Embeddings & Positional Encoding

How the model understands text

- **Tokenisation** → breaks text into units
- **Embeddings** → converts each token into a high-dimensional vector (similar words cluster together)
- **Positional Encoding** → adds order information (models don't understand sequence naturally)

Result: Text becomes a mathematical structure the neural network can process

Transformer Architecture – The 2017 Breakthrough

“Attention Is All You Need”

- Replaced slow, sequential **RNNs/LSTMs** with **parallel processing**
- Core innovation: **Self-attention** mechanism
 - Each token attends to every other token
 - Captures long-range dependencies and context effectively

Benefits:

- Much faster training on GPUs
- Superior coherence in long documents
- Enabled the rapid scaling of modern LLMs

Training Objective & Inference Loop

Two distinct phases:

Training (Self-supervised learning):

- Next-token prediction on billions of text examples
- Adjusts billions of parameters via backpropagation
- Example: Google Translate quality jumped >50% after switching to Transformers

Inference:

- Autoregressive loop – generates tokens one by one in real time
- No further learning; parameters are fixed
- **Temperature** controls randomness vs consistency

Why Bigger Models Suddenly “Reason”

Scaling Laws & Emergent Abilities:

Performance improves predictably with:

- More parameters
- More training data
- More compute

Emergent abilities: Certain capabilities (multi-step reasoning, long coherence, complex instruction following) appear suddenly at scale

Smaller models: Good for simple tasks

Larger (frontier) models: Excel at sustained reasoning and structured outputs

Frontier Models & Strategic Considerations (2025–2026)

Model Selection as a Strategic Decision:

- **Closed models** (OpenAI, Anthropic, Google): Superior out-of-the-box capability, ease of use
- **Open models** (Meta, Mistral): Greater control, data residency, customisation

Key Evaluation Criteria:

- Real-world workflow performance (not just benchmarks)
- Cost & latency at scale
- Privacy, compliance, and governance
- Multimodal capabilities (text + image + audio)

Final takeaway: Choose models that align with your specific business needs, risk tolerance, and operational constraints.

Summary – Key Takeaways

- **Generation** = creating usable business content (not predicting outcomes)
- Built on **autoregressive next-token prediction**
- Powered by **Transformer architecture + self-attention**
- Learns via massive **next-token prediction** training
- Performance scales with size → emergent reasoning abilities
- **Always probabilistic** → human oversight is non-negotiable
- Success depends on prompt quality, token management, and responsible deployment

Remember: Generative AI is not magic — it is architecture, data, scale, and probability working together.